PANJABUNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. Under the Panjab University Act VII of 1947 –enacted by the Govt. of India)

SYLLABI

Course Outline for
B.Voc (Food Science and Technology)

as per NEP 2020

w.e.f. session 2024-25

Course Outline for B.Voc Food science and Technology as per NEP 2020 w.e.f. session 2024-25

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FST 101	Core Discipline	Introduction To Food Microbiology	2	2	4	40	10	40	10	100
FST 102	Core Discipline	Bakery And Confectionery Technology-I	2	2	4	40	10	40	10	100
FST 103	Core Discipline	Bakery And Confectionery Technology-II	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 101	Multidisciplinary Course	Fundamentals of Information Technology[2]	3	-	3	60	15	-	-	75
SEC I	Skill Enhancement Course	SEC Options[1]			3					75
VAC I	Value-Added Course	VAC Options[3]			2					50
PW I	Project Work	Project Work-I& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FST 201	Core Discipline	Dairy Technology	2	2	4	40	10	40	10	100
FST 202	Core Discipline	Milk And Dairy Food Products	2	2	4	40	10	40	10	100
FST 203	Core Discipline	Management of Food Industry	4	-	4	80	20	-	-	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 201	Multidisciplinary Course	Soft Skills and Personality Development[2]	3	-	3	60	15	-	-	75
SEC II	Skill Enhancement Course	SEC Options[1]			3					75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role – Baking Technician, Dairy Products Processor

*Summer Industrial Training of 2-4 weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 3rd Semester. Viva-Voce examination to be held within 3 weeks of the start of 3rd semester.

** Minimum Credits required to get Certificate Course in Food Science and Technology = 48 + 4 #.

#Exit 1: Certificate Course-Student can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

ESSENTIAL COMPONENTS OF COURSE OF B.Voc Food science and Technology (SEMESTERS I & II)

PROGRAMME OUTCOMES

PO1. They will be able to demonstrate knowledge of Food Chemistry, Food Microbiology, Food science and Food engineering.

PO3. They will demonstrate an ability to identify, formulate and solve processing problems related to food industry

PO4. They will able to focus on the importance of safe processed nutritious food.

PO5. They will get exposure to various technologies aspect of Food product preparation.

PO6. They will demonstrate an ability to design or process food products as per the needs and specifications.

PO7. They will demonstrate an ability to work in Food industries, research organization and teaching.

Semester I

	COURSE TITLE	INTRODUCTION TO FOOD MICROBIOLOGY
1.	**Course Code**	FST 101
2.	**Typology of Course**	Core Discipline :
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100 Marks (80TH/PR+20IA**)** i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Baking Technician
7.	**Course Objectives**	To understand the types of food microbes, causes for food spoilage, process for food spoilage, need for food preservation.
8.	**Course Outcomes**	On successful completion of the course, the Students will get a basic knowledge about significance of microbes in different types of food, microbial growth characteristics, various methods of food preservation, hurdle technology etc.
9.	**Pedagogy of the Course Work**	Class Room, Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT I

Introduction to Food Microbiology

• Definition and Scope of food microbiology
• Characteristics of Microorganisms in Food
• Types of microorganisms, Classification and Nomenclature, Morphology and Structure and their importance in food (bacteria, fungi, viruses and prions, protozoan and others)
• Significance of spores
• Heat resistance of microorganisms and their spores –bacteria and their spores – fungi and their spores –yeast and their spores - spoilage of canned foods and types of spoiled cans

UNIT II

• Microbial Growth in Food
• Microbial Growth Characteristics- Bacterial growth curve, microbial reproduction and microbial growth in food
• Factors affecting the growth of microorganisms in food

UNIT III

Microbial Food Spoilage

• Sources of Microorganisms in foods
• Some important food spoilage bacteria
• Food spoilage: Microbial, physical, chemical & miscellaneous.
• Changes caused by micro-organisms during spoilage (breakdown of proteins, carbohydrates, fats and other constituents)
• Detection of food pathogens: Overview of Conventional and Rapid methods to detect food pathogens.

UNIT IV

Control of Microorganisms in Foods

• Principles and methods of preservation
• Physical Methods of Food Preservation- Dehydration, Freezing, Cool Storage, Heat Treatment (esp. thermo bacteriology), Irradiation
• Chemical Preservatives
• Bio preservatives
• New Non Thermal methods
• Introduction to Hurdle concept and Predictive Microbiology

Practical

1. Introduction to the Basic Microbiology Laboratory Practices and Equipments
2. Functioning and use of compound microscope
3. Cleaning and sterilization of glassware
4. Preparation and sterilization of nutrient broth
5. Cultivation and sub-culturing of microbes
6. Preparation of slant, stab and plates using nutrient agar
7. Morphological study of bacteria and fungi using permanent slides
8. Methods of sterilization and preparation of media
9. Simple staining
10. Endospore staining
11. Standard Plate Count Method

12. Suggested Resources for Reading

1) Frazier William C and Westhoff, Dennis C. Food Microbiology, TMH, New Delhi, 2004

2) Jay, James M. Modern Food Microbiology, CBS Publication, New Delhi, 2000

3) Garbutt, John. Essentials of Food Microbiology, Arnold, London, 1997

4) Pelczar MJ, Chan E.C.S and Krieg, Noel R. Microbiology, 5th Ed., TMH, New Delhi, 199

5) Adams M.R and Moss M.O, 2004 "Food Microbiology", Panima Publishing corporation, New Delhi, 2ndEdition.

6) Sivasankar B, 2009. "Food Processing and Preservation", PHI Learning Private Limited, Eastern Economy Edition, 6th edition,

7) William C Frazier and Dennis C. Westoff , 2008. "Food Microbiology", Special Edition, Springer, The Mc Graw-Hill Companies.

Note: Latest edition of text books may be used**.**

	COURSE TITLE	BAKERY AND CONFECTIONERY TECHNOLOGY-I
1.	**Course Code**	FST102
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100Marks (80TH/PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Baking Technician
7.	**Course Objectives**	i) To provide know about the machinery and process involved in the baking process ii) To understand the various types of sugar and its grades iii) To know the confectionery product manufacture
8.	**Course Outcomes**	On successful completion of the course, the Students will be able to CO1. Demonstrate a comprehensive understanding of the machinery, processes, and techniques involved in bakery and confectionery production, encompassing ingredient selection, mixing methods, baking temperatures, and quality control measures. CO2. Develop technical skills in operating bakery and confectionery equipment, conducting quality tests on raw materials and finished products, and implementing hygiene and sanitation practices to ensure food safety and product quality
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT I

- Introduction to food processing and preservation.
- Various products of the bread and bakery sub-sector
- List the various types of industries within the bakery sub-sector
- Explain the baking process
- Equipments used in Bakery (Dough mixers, Dividers, rounders, Proofing, molding, Ovens, Slicers etc), Cleaning and maintenance of the work area and machineries, Baking ingredients required for production and plan production sequence
- Testing of Flour for Bakery Goods: Laboratory testing of Wheat grain Quality, Moisture tests, Grain hardness testing. Testing, Visco graph, Amylograph, Ferinograph. Units of measurement used in the baking industry.
- Raw material required for bakery products. Role of flour, water, yeast, salt, Sugar milk and fats.
- Yeast- An elementary knowledge of Baker's yeast, the part it plays in th fermentation of dough and conditions influencing it's working.
- Effect of over and under fermentation and under proofing of dough and other fermented goods.

UNIT II

- Process of mixing and knead ingredients to make dough
- Oven & Baking – Knowledge and working of various types of oven.
- Biscuit-Types of biscuit dough – Developed dough, short dough, semi-sweet, enzyme modified dough and batters- importance of the consistency of the dough, Factors affecting the quality of biscuits/ cookies
- Cake – Flour specification – ingredients: Cake making ingredients- flour, sugar, shortening and egg types of chemically aerated goods Moistening agents Fats and Oil leavening agents
- Manufacturing process – Cake making methods sugar batter process, flour batter process, genoise method and blending and rubbing method, Correct temperature for baking different varieties of cakes, Balancing of cake formula Characteristics of cakes: external: Internal ,Types of icing.

UNIT III

- Types of Confectionery, raw materials and processing of chocolates, hard boiled candies.
- Additives for Confectioneries.
- Equipments used in Confectionery manufacture
- Toffees – Milk toffee; chocolate; stick jaws; liquor chocolate.
- Baking temperatures for confectionery goods.
- Process of proofing process of baking products in the oven
- Process of cooling of baked products.

UNIT IV

Bread manufacturing process – Straight dough fermentation Bread improvers- improving physical quality

Methods of bread making

- Straight dough methods
- Delayed salt method
- No time dough method
- Sponge and dough method Characteristics of good bread
- External Characteristics – volume, symmetry of shape
- Internal characteristics- color, texture, aroma, clarity and elasticity Bread faults and their remedies.
- Bread diseases- Rope and Mould – causes and prevention.

Practical

1. Preparation of cookies and biscuits
2. Preparation of different types of cakes
3. Preparation of different types of pastries
4. Preparation of different types of ice creams (Vanilla, Strawberry, chocolate, Pineapple, Mango)
5. Preparation different types of Pudding
6. Preparation of Bread rolls; Bread sticks & softs rolls;
7. Preparation of Fermented doughnuts;
8. 10. Preparation of Pizza base

12. Suggested Resources for Reading

1. Samuel A. Matz, "Bakery Technology and Engineering", Chapman & Hall, 3rd Edition, 1992.
2. Reference Books 1. "Association of Operative Millers Cereal Millers Hand Book", Burgess Publishing company, USA, 1963.
3. Pomeranz Y, "Modern Cereal science and Technology" MVCH Publications, NY, 1987.
4. Kent N.L .Evers A.D, "Technology of Cereals" Peregaman Press, Elsevier Publishers, 1994.
5. Samuel A. Matz, "Equipment for Bakers" Pan Tech International Publication, 1988.
6. Stanley P Cauvain, Linda S Young, "Technology of Bread making", Aspen publication, 2ndEdition, 2007.
7. "Association of Operative Millers Cereal Millers Hand Book", Burgess Publishing company, USA, 1963.
8. Kent N.L.Evers A.D, "Technology of Cereals" Peregaman Press, Elsevier Publishers, 1994.

Note: Latest edition of text books may be used**.**

	COURSE TITLE	BAKERY AND CONFECTIONERY TECHNOLOGY-II
1.	**Course Code**	FST 103
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100Marks (80TH/PR+20IA) i) The exam of this course will be of 3 hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Baking Technician
7.	**Course Objectives**	To understand the various types of sugar and its grades To know the confectionery product manufacture
8.	**Course Outcomes**	On successful completion of the course, the Students will be able to CO1. Acquire problem-solving abilities to tackle challenges related to ingredient sourcing, production planning, equipment maintenance, and quality assurance in bakery and confectionery operations. CO2. Demonstrating professional responsibility, they will adhere to industry standards, regulations, and best practices, including compliance with food safety laws, hygiene standards, and waste management protocols. CO3.To provide students' knowledge about the machinery and process involved in the baking process
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

Unit I

- Bakery layout- The required approvals for setting up of a Bakery- Government procedure and Bye laws.
- Selection of site & Selection of equipment
- Layout design
- Electricity
- Process of cleaning the work area and machineries after production
- Roles and responsibilities of a bakery production specialist.
- Plan, organize and management of work in bakery.
- Method of documenting and recording the details of baking ingredients to final finished product

Unit II

- Quality control of raw material and of finished products
- Role of Hygiene in Bakery
- Personal hygiene, care of skin, hand and feet, Food handlers hygiene, protective clothing.
- Dishwashing methods- manual and machine dish washing- merits and demerits
- Garbage disposal- different methods- advantages and disadvantages
- Food poisoning- Causative factors and the precautions to be taken by food handlers.

Unit III

- Food Storage- Techniques of correct storage, storage temperature of different commodities prevent bacterial manifestation or contamination. Storage of confectionery products.
- Pest Control- Rodents and insect control techniques, special stress on control of flies , rats and cockroaches, care of the premises and equipment
- Municipal health laws.
- Golden rules of first aid and treatment for cuts, wounds, burns
- Process of packaging of baked products

Unit IV

- Accounts and costing:
- Purchase book, purchase return book, Stores requisition.
- Sales book, Sales return book, Cash voucher. Credit voucher book
- Preparation of invoices and debit/ credit memos
- Percentage and discounts
- Introduction to ingredient costs, labour costs, overheads, gross profit calculation of cost price, sales price and catalogue price.
- Commodities: Sugar, Cocoa, Chocolate, Milk, Butter, Cream, Cheese, Food colours, Flavors & essences
- Dry fruits & nuts used in confectionery, Fresh and preserved fruit products
- Food laws- Agmark

Practical Work:-

- Prepare the Bread rolls/cakes/ and assessment of its quality
- Perform the quality test on fat/butter for bakery applications on different parameters
- Carry out the assessment of market Candy and Chewing gum.
- Perform the quality assessment test on yeast and skimmed milk powder for bakery application.
- Prepare Chocolate cookies and assessment of its quality.

12. Suggested Resources for Reading

1. Samuel A. Matz, “Bakery Technology and Engineering”, Chapman & Hall, 3rd Edition, 1992.
2. Reference Books 1. “Association of Operative Millers Cereal Millers Hand Book”, Burgess Publishing company, USA, 1963.
3. Pomeranz Y, “Modern Cereal science and Technology” MVCH Publications, NY, 1987.
4. Kent N.L.,.Evers A.D, “Technology of Cereals” Peregaman Press, Elsevier Publishers, 1994.
5. Samuel A. Matz, “Equipment for Bakers” Pan Tech International Publication, 1988.
6. Stanley P Cauvain, Linda S Young, “Technology of Bread making”, Aspen publication, 2ndEdition, 2007.
7. “Association of Operative Millers Cereal Millers Hand Book”, Burgess Publishing company, USA, 1963.
8. Kent N.L.,.Evers A.D, “Technology of Cereals” Peregaman Press, Elsevier Publishers, 1994.

Note: Latest edition of text books may be used.

Semester- II

	COURSE TITLE	DAIRY TECHNOLOGY
1.	**Course Code**	FST 201
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100Marks (80TH/PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Dairy Products Processor
7.	**Course Objectives**	To know the compositional and technological aspects of milk To study processed milk products
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1. To understand the technology and constituents of dairy products and their preparation CO2. To develop technical proficiency in various areas such as milk composition analysis, dairy processing techniques, quality control measures, and sanitation protocols in dairy industry settings. Additionally, students will acquire problem-solving skills to identify, analyze, and address challenges related to milk processing, product development, quality assurance, and food safety in dairy operations.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT I

TECHNOLOGY OF MILK AND MILK PRODUCTS

Composition and Physical properties of milk , milk color, taste, pH and buffering capacity, refractive index, viscosity, surface tension, freezing, boiling point, specific heat, OR, electrical conductivity.

UNIT II

Milk fat

• Composition and structure, factors affecting melting point, boiling point, solubility and Refractive Index, fat constants (saponification value, iodine value, RM value, Polenske value, peroxide value).

• Chemical reactions of fat (hydrolysis, auto-oxidation), condition favoring auto oxidation, prevention, measurement of auto-oxidation.

UNIT III

Fats and Oils

• Classification of lipids, types of fatty acids - saturated fatty acids, unsaturated fatty acids, essential fatty acids, Trans fatty acids. (Ch-3, Manay)

• Rancidity - hydrolytic and oxidative rancidity and its prevention.

• Define - margarine, butter, hydrogenated vegetable oil, lard.

Protein and Enzymes

• General structure, amphoteric nature, difference between casein and serum protein, different types of casein (acid and rennet), uses of casein, fractionation of protein.

• Enzymes- catalase, alkaline phosphatase, lipases and proteases.

UNIT IV

• Liquid milk handling operations: Systems of collection of milk, Reception, Platform testing, Process of testing milk for accepted quality standards, organoleptic test of milk, Procedure for COB test of milk

• Various stages of processing: Filtration and Clarification, Skimming, Standardization Homogenization, Pasteurization (LTHT, HTST, UHT) Sterilization, Packaging, Storage and distribution of fluid milk

• Description and working of clarifier, cream separator, homogenizer and plate heat exchanger. Need for processing milk.

• List the various units within a dairy processing plant

• List the machineries used in a dairy processing plant

• Roles and responsibilities of a dairy products processor in a dairy processing plant

• Flow diagram of following milk products – Butter, ghee, flavored milk, yoghurt, dahi, shrikhand, ice-cream, condensed milk, milk powder, channa, paneer, cheese (cheddar)

• State the composition and nutritive value of the milk products

Practical Work: -

1. To perform platform tests in milk.(Acidity, COB, MBRT ,specific gravity, SNF)
2. To estimate milk protein by Folin method.
3. To estimate milk fat by Gerber method.
4. Preparation of flavoured milk/. Pasteurization of milk
5. To prepare casein and calculate its yield.

12. Suggested Resources for Reading

1. De Sukumar, Outlines of Dairy Technology, Oxford University Press, Oxford. 2007
2. Hall GM, Fish Processing Technology, VCH Publishers Inc., NY, 1992
3. Sen DP, Advances in Fish Processing Technology, Allied Publishers Pvt. Limited 2005
4. Shahidi F and Botta JR, Seafoods: Chemistry, Processing, Technology and Quality, Blackie Academic & Professional,London,1994
5. Webb and Johnson, Fundamentals of Dairy Chemistry

Note: Latest edition of text books may be used.

	COURSE TITLE	MILK AND DAIRY FOOD PRODUCTS DEVELOPMENT
1.	**Course Code**	FST 202
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100Marks (80TH/PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Dairy Products Processor
7.	**Course Objectives**	To enable the students to understand the various dairy products and their preparation.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1. To apply theoretical knowledge and practical skills acquired during the course to optimize milk handling operations, improve product quality, and enhance efficiency in dairy processing plants, explore new concepts, technologies, and processes in dairy industry advancements, ethical conduct in dairy industry practices, including adherence to hygiene, sanitation, safety guidelines, and responsible waste management. CO2. To make the students aware about the diverse milk products and practically capable to develop new products for the market CO3. To demonstrate a comprehensive understanding of the compositional and technological aspects of milk, dairy products, and their development processes.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

Unit I

• Dairy industry in India & its future prospects.
• Process of production planning
• State the factors affecting operation efficiency during production
• New Concept in milk processing- UHT, Membrane processing, Microwave and irradiation treatments; aseptic packaging. Microbiology of milk, sources of milk contamination.

Unit II

• Process for producing dairy products and process of pasteurization :
• Ingredients required for production
• State the production process of pasteurization
• Process of separation and bactofugation
• Method of standardization of milk
• Method of homogenization of milk
• Method of heat exchange during pasteurization
• Process of HTST pasteurization
• Process of producing lassi, flavoured drinks, cheese dahi, kalakand, ice-cream, cream, butter , cooking butter, ghee

Unit III

• Technology of condensed and evaporated milk: Composition, nutritive value, process of manufacture, defects (their causes and prevention).
• Technology of yoghurt, Acidophilus milk, Bufgaricus milk, Kumiss, Kefir; Manufacturing of Cheddar, Mozzarella, cottage and processed cheese.
• Milk adulteration and quality control in dairy industry. By products of dairy industry and their utilization.
• Milk based infant foods; Manufacturing of casein, Caseinate, Co-precipitates, WPC, Lactose; National and International Organizations in dairy Industry.
• Milk and milk product standards and legislations in India: Grading of milk and criterion of grading, reconstituted milk, synthetic milk.
• List the different packaging materials used to pack dairy products
• Method of packaging dairy products

Unit IV

• Methods of cleaning and sanitization: Cleaning of production area, equipment, and tools used Equipment, detergents and sanitizers used in the cleaning and maintenance of the work area, Properties of the cleaning agents used, CIP method of cleaning,
• Lubrication system and its process followed in the dairy industry
• State the different types of maintenance procedures, Periodic maintenance of all production machineries
• Method of managing and disposing waste material
• Personal hygiene and sanitation guidelines
• Demonstrate how to use tools safely
• Food safety hygiene and quality standards to follow in a work environment , HACCP principles to eliminate food safety hazards in the process and products
• Method of documenting and recording the details of raw material to final finished product

Practical Work: -

1 Platform tests,
2 Determination of fat, SNF, TS Protein,
3 Lactose and ash contents of milk.
4 Layoput plan for setting up of milk plant,
5 Preparation and Evaluation of different types of milk and milk products 6
Preparation and evaluation of butter, ice cream, cheese, yoghurt

12. Suggested Resources for Reading

1. Sukumar. 1983. Outlines of Dairy Technology. De,Oxford University Press.
2. James N.Warner.1989. Principles of Dairy Processing by, Wiley Eastern Ltd.
3. Eckles,Combs; and Macy. 1982. Milk and Milk Products by,Tata McGraw Hill.
4. Aneja et al. 2002.Technology of Indian Milk Products.A Dairy India Publication.

Note: Latest edition of text books may be used.

	COURSE TITLE	MANAGEMENT OF FOOD INDUSTRY
1.	**Course Code**	FST 203
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	60 hours of lectures for Theory (1 credit = 15 hours of theory learning based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits
5.	**Max. Marks and Time**	100 Marks (80TH/PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Dairy Products Processor
7.	**Course Objectives**	i) To develop a knowledge base in key areas of institutional food management. ii) To impart necessary expertise to run a food service unit. iii) To provide practical level experience in managing food service management. iv) To critically evaluate the functioning of food service units.
8.	**Course Outcomes**	On successful completion of the course, the Students will be able to CO1. To impart necessary expertise to run a food service unit. CO2. To critically evaluates the functioning of food service units. CO3 To provide practical level experience in managing food service management
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.
11.	**Course Content**	**Unit-I**

• Institutional management, Levels of Management.
• Review of different types of food service operations. Commercial, non- commercial food service institutions.
• Importance of food standards. Need for knowledge as to what constitutes good products, criteria for good quality products, hazards of poor quality, quality control, Specification for quality in food products.
• Organization - definition and types

Unit-II

•The Manufacturing (Production) Function: Objectives of Production Management, Operation Concept, Meaning of Product Development, Factors Influencing Choice of Manufacturing Systems, Classification of Manufacturing Systems;
• Basic Procedure of Method Study (Work Study) and Time Study; Concepts, Objectives and functions of Production Planning and Control (PPC).
• Physical plant- its location, floor plans, space allowance, Institutional food units-storage, preparation and serving
Cost control
a) Food cost, Labour cost, Maintenance cost
b) Budgets
c) Documentation and Records: Documentation and maintenance of record of raw materials and packaging materials
d) Portion control
• Role of supervisor in fruits and vegetable processing industry. • Planning and organization of work: organization standards, process standards and procedures followed in the organization, types of products produced by the organization, Code of business conduct, Dress code.

Unit-III

• Personnel Management
• Personnel and leadership qualities for food service administration.
• Labour, Types of labour, criteria for selection and employees training.
• Labour laws and legal aspects- health & safety of employees, welfare policies.
• Manage and lead a team Management Skills-The Importance of Delegation, Developing Your Team, Motivating Your Team: Motivation Theories – Maslow's, Herzberg's; Worker's Participation in Management; Collective Bargaining.
• Communicating and Working With Your Team – and With Others, Communicating With People Outside Your Team
• Managing Discipline

Unit-IV

Hygiene and Sanitation
i) Personal hygiene- Importance of personal cleanliness in handling and serving food. Health examination of personnel. ii) Sanitation in handling food and equipment.
a. Preventing contamination of cooked food and handling of fresh foods,. b. Cleaning and hygienic handling of equipments. c. Insect and Rodent control d. Safety, general safety rules in food preparation and service area. Accident prevention
Equipment- Types of equipment- criteria for selection, operation and care.
Food laws and regulations on raw materials, product(s), packaging and labelling
Food safety, hygiene, sanitation GMP , HACCP and quality assurance procedures for disposal of waste.

12. Suggested Resources for Reading

1 West Wood A; Harper Food Service in Institution.

2. West, Bessin, Brooks; Food Service in Institutions
3. A.M. Home Economics Association; Hand Book of Food Preparations:
4. Sweetman, M.M. 4, Mac, Keller; Food Selection and Preparation:
5. Oliver B., Watson; School Lunch Room Service
6. Lender H. Katshever and Margret E. Terrel; Food Service Planning: Layout Equipment
7. Davidson and Passmore- Human Nutrition and Dietetics.

Note: Latest edition of text books may be used**.**